

Optimization Study of SVM Classification in a Synthetic PA Cohort

SVMs: Hard-Margin Feasibility, Soft Margins, and Kernel Validation

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How the analysis is organized

hard margin → check feasibility → soft margin → RBF comparison →
model choice

- Start with a **hard-margin SVM**, which allows no classification errors.
- Check whether its constraints are feasible in three different ways.
- Move to a **soft-margin SVM** once the hard-margin model is ruled out.
- Compare the linear model with an **RBF-kernel SVM**.
- Choose the model using nested cross-validation.

Analysis plan

- 1 Test the hard-margin model.** It requires a linear boundary that classifies every observation correctly with a margin. We first ask whether those constraints are feasible.
- 2 Confirm the infeasibility result.** Three checks show that the classes are not linearly separable: (2a) the LP solver returns “infeasible”; (2b) a 12-patient Farkas certificate turns that into a proof ($0 \geq 1$); (2c) the two convex hulls overlap.
- 3 Fit a soft-margin model.** Slack variables allow margin violations, controlled by the penalty C . We tune C , inspect the fitted boundary, and also solve the primal with stochastic subgradients.
- 4 Compare with an RBF kernel.** The kernel allows a nonlinear boundary. Nested cross-validation tests whether that added flexibility improves performance.

Key point: The hard-margin analysis is a feasibility check; model performance is assessed only after moving to soft margins.

The motivating classification problem

Primary aldosteronism (PA) is a common, *curable* cause of high blood pressure.

- **Unilateral (UPA)**: one gland — often *cured by surgery*.
- **Bilateral (BPA)**: both glands — managed with *medication*.

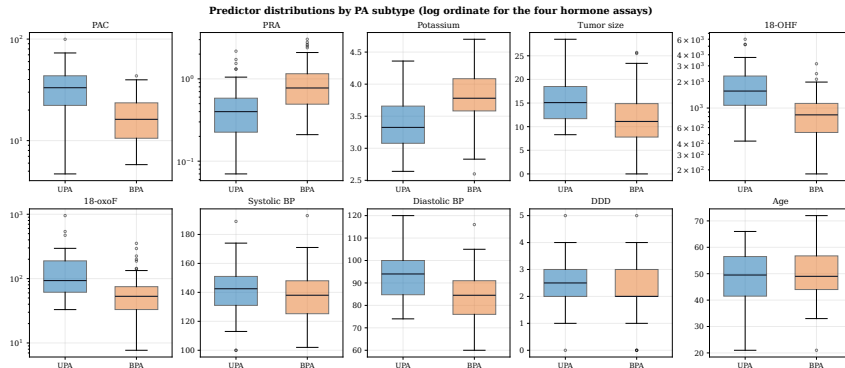
Getting the subtype right changes treatment. Here we study the optimization workflow on synthetic data; this is not evidence of clinical utility.

Key point: The task is to distinguish UPA from BPA using ten routinely measured predictors.

The dataset

- 114 synthetic patients
- 56 UPA / 58 BPA
- 10 numerical predictors (PAC, PRA, potassium, tumor size, steroids, BP, ...)

What the data look like before any model



114 patients, no missing values. **UPA** versus **BPA**; hormone assays on a log ordinate.

Key point: Every predictor overlaps across the two groups, so no single measurement provides a clean split.

Descriptive statistics — the subtype contrast

Lecture ref: Statistical Learning Models (W9)

Feature	UPA	BPA	Cohen's d	Welch t	p
PAC (ng/dL)	34.9	18.4	1.17	6.20	<0.001
18-OHF (ng/dL)	1883	942	0.99	5.25	<0.001
Potassium (mmol/L)	3.38	3.79	-0.95	-5.09	<0.001
PRA (ng/mL/h)	0.51	0.96	-0.80	-4.32	<0.001
Diastolic BP (mmHg)	92.3	84.1	0.75	4.03	<0.001
Tumor size (mm)	15.4	11.4	0.74	3.98	<0.001
18-oxoF (ng/dL)	148.1	71.8	0.66	3.46	<0.001
DDD (drugs)	2.48	2.14	0.36	1.94	0.055
Systolic BP (mmHg)	141.4	137.5	0.23	1.22	0.224
Age (years)	48.6	50.1	-0.15	-0.80	0.425

Bold: $p < 0.05$ (Welch t -test), indicating a difference in group means (7 of 10). Sign: + means higher in UPA.

Cohen's d — how big

Gap between *class means*, in pooled SDs: 0.2 small, 0.5 medium, 0.8+ large.

Welch t — how certain

Same gap $\div$ standard error; grows with $\sqrt{n}$. Evidence, not size.

Key point: Several predictors differ between groups, but their distributions still overlap.

Can a hard-margin SVM fit these data?

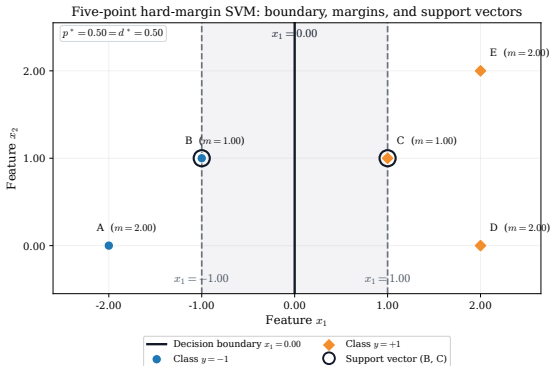
The hard-margin SVM maximizes the distance between the two supporting hyperplanes:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{s.t.} \quad t_i(w^\top x_i + b) \geq 1.$$

- Score $s = w^\top x + b$; sign is the prediction.
- Margin width $= 2/\|w\|$.
- **Zero mistakes allowed.**

This is a convex quadratic program, provided the constraints are feasible.

Lecture ref: *Statistical Learning Models (W9)*; *Convex Optimization Models (W8)*



Five-point teaching example: boundary, margins, support vectors.

Three checks for linear separability

Lecture ref: Introduction to Duality (W9); First-Order Optimality (W10)

We check the hard-margin constraints in **three different ways**. All three reach the same conclusion.

2a. Ask the solver

Run the feasibility LP on all data and every fold.

2b. Farkas certificate

Use 12 nonzero weights to derive the contradiction $0 \geq 1$.

2c. Convex hulls

Show that the two class hulls share a point.

The solver gives the numerical result, while the certificate and convex hulls verify it algebraically and geometrically.

The solver finds no feasible hard margin

Feasibility of the hard-margin constraints is itself a **linear program**: does any (w, b) satisfy

$$t_i(w^\top x_i + b) \geq 1 \quad \text{for all } i?$$

- HiGHS returns **status: infeasible** on the full 114-patient dataset.
- Repeated inside every one of the 5 cross-validation folds — **all infeasible**.
- So the result is not an artifact of one particular split.

Key point: The hard-margin constraints are infeasible on the full dataset and in every training fold.

Solver verdict

Subset	Feasible?
Full data (114)	No
Fold 1	No
Fold 2	No
Fold 3	No
Fold 4	No
Fold 5	No

A Farkas certificate confirms infeasibility

Farkas' lemma: exactly *one* of two systems can hold — either a separating w exists, **or** a set of nonnegative weights λ gives a contradiction.

- The certificate has nonzero weights for **12 observations**, with $\lambda_i \geq 0$.
- Combining their constraints gives $0 \geq 1$, which is impossible.
- The calculation can be checked independently of the solver status.

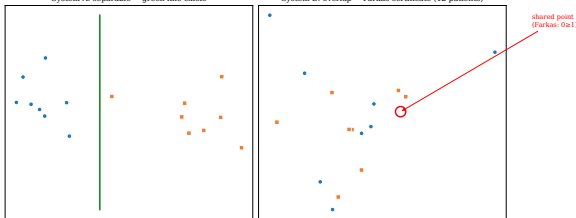
Key point: The certificate proves that the hard-margin constraints are inconsistent.

Lecture ref: *Introduction to Duality (W9) — theorem of the alternative*

Farkas' lemma: exactly one system is solvable

System A: separable → green line exists

System B: overlap → Farkas certificate (12 patients)



Exactly one system can be feasible. For these data, it is the certificate system on the right.

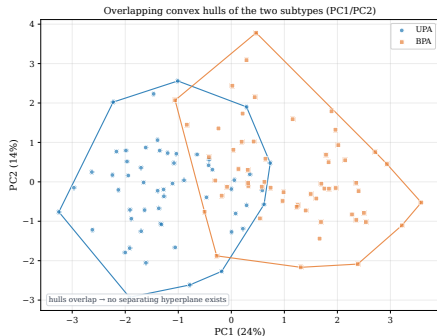
The convex hulls explain the infeasibility

After rescaling the same λ weights, we obtain **the same point from both class convex hulls**:

$$\sum_i \beta_i^+ x_i = \sum_i \beta_i^- x_i,$$

matching to $< 10^{-15}$ (machine precision).

- A shared point $\Rightarrow$ no hyperplane separates the classes.
- This gives a geometric interpretation of the solver and Farkas results.



The synthetic cohort in a 2-D projection. The exact certificate is computed in the full feature space.

Soft margins allow violations

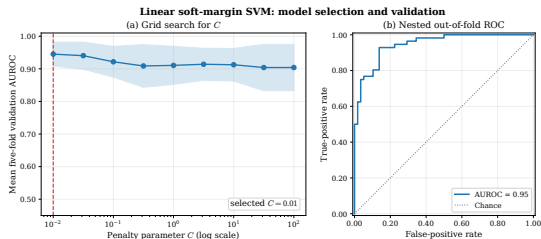
Slack variables ξ_i allow observations to fall inside the margin or be misclassified:

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$$

$$\text{s.t. } t_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

- C controls the penalty assigned to margin violations.
- Nested cross-validation selected $C = 0.01$.
- The model can now accommodate overlap between the classes.

Lecture ref: *Statistical Learning Models (W9)* — hinge loss, slack, penalty C



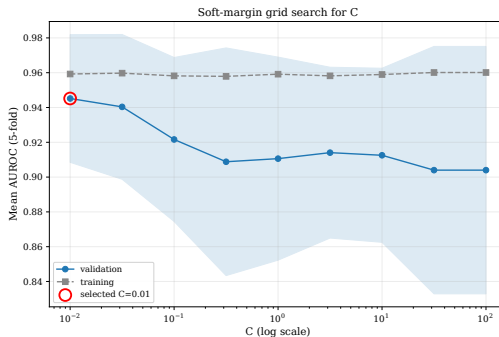
(a) validation AUROC vs C ; (b) nested out-of-fold ROC, seed-zero AUROC 0.95; 20-seed mean 0.94.

Cross-validation selects $C = 0.01$

For **each candidate** C and inner split:

- fit preprocessing on the training fold;
- solve that fixed- C SVM quadratic program to obtain (w_C^*, b_C^*) ;
- score the held-out validation fold.

We average validation AUROC over five folds and choose its maximum. The untouched outer fold is not used for tuning. Validation AUROC peaks at $C = 0.01$ and declines as C increases.



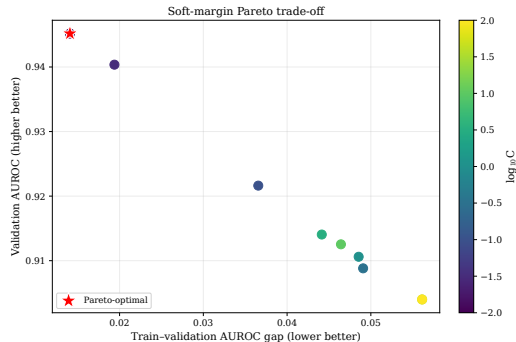
Mean validation vs training AUROC across the C grid; band is ± 1 SD over folds. The selected $C = 0.01$ is circled.

Key point: Within the prespecified grid, $C = 0.01$ gives the best validation AUROC.

$C = 0.01$ also has the smallest generalization gap

Each value of C is plotted by its train-validation gap and validation AUROC. Preferred settings lie near the **top-left**.

- $C = 0.01$ is the only **Pareto-optimal** point in this grid.
- Every larger value has lower validation AUROC and a wider gap.



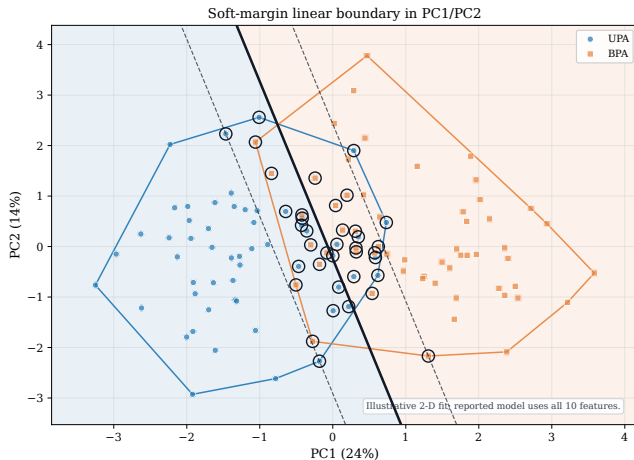
Color shows $\log_{10} C$; the star marks the selected model.

Key point: $C = 0.01$ has both the highest validation AUROC and the smallest train-validation gap.

Where the soft-margin boundary falls

The first two principal components provide a 2-D view of the fitted **soft-margin boundary**.

- The boundary passes through the region where the two class hulls overlap.
- Circled observations are **support vectors**, concentrated near the boundary.
- Slack variables account for observations inside the margin or on the wrong side.



Solid: decision boundary; dashed: the ± 1 margins. Hulls overlap, so no line separates them cleanly.

From the constrained SVM to a subgradient objective

Lecture ref: *Statistical Learning Models (W9); The Stochastic Gradient Method (W10)*

1. Start with the constrained soft-margin problem

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i, \quad \begin{aligned} t_i(w^\top x_i + b) &\geq 1 - \xi_i, \\ \xi_i &\geq 0. \end{aligned}$$

The smallest feasible slack is

$$\xi_i = \max\left(0, 1 - t_i(w^\top x_i + b)\right),$$

which is the **hinge loss** $\ell_i(w, b)$.

2. Substitute the hinge loss and rescale

$$\frac{1}{2} \|w\|^2 + C \sum_i \ell_i(w, b).$$

Dividing by the positive constant mC does not change the minimizer:

$$P(w, b) = \frac{1}{m} \sum_i \ell_i(w, b) + \frac{\lambda}{2} \|w\|^2, \quad \lambda = \frac{1}{mC}.$$

A stochastic method can therefore sample one observation and use its subgradient at each step.

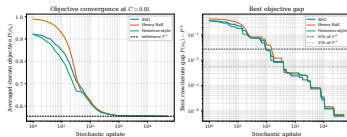
Key point: Because hinge loss is nondifferentiable at margin one, the Pegasos update is a stochastic subgradient step.

After grid search, optimize only the selected $C = 0.01$

Lecture ref: *The Stochastic Gradient Method; Momentum Methods (W11)*

This is not another C search. Every method solves the one primal selected above: $C = 0.01$, hence $\lambda = 1/(114C) = 0.8772$. They use the same zero start and 50,000-sample sequence, with $\eta_k = a/[\lambda(k + 100)]$.

Method	5% steps	1% steps	$P(\bar{w}_K)$	AUROC
SSG	96	389	0.5544	0.959
Heavy Ball	150	391	0.5544	0.959
Nesterov-style	98	389	0.5544	0.959



Key point: Across ten new sampling sequences, momentum has no robust advantage; smooth acceleration guarantees do not apply to hinge loss.

An RBF kernel allows a nonlinear boundary

Lecture ref: Introduction to Duality (W9); Statistical Learning Models (W9) — kernels

1. Start from the soft-margin primal:

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i \quad \text{s.t.} \quad t_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

2. In the Lagrangian dual, the data enter **only through inner products** $x_i^\top x_j$:

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j t_i t_j \underbrace{x_i^\top x_j}_{\text{inner product}} \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i t_i = 0.$$

3. Replace the inner product. Map $x \mapsto \varphi(x)$ and use a **kernel**:

$$x_i^\top x_j \longrightarrow K(x_i, x_j) = \varphi(x_i)^\top \varphi(x_j).$$

4. For the RBF kernel:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2).$$

This gives a nonlinear boundary in the original space while the dual remains a convex QP.

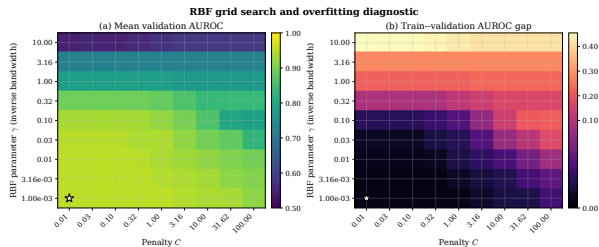
$$\text{Decision score becomes } f(x) = \sum_i \alpha_i t_i K(x_i, x) + b.$$

RBF tuning favors the smoothest boundary

The RBF kernel $K(x_i, x_j) = e^{-\gamma \|x_i - x_j\|^2}$ can bend the boundary.

- Nested cross-validation tested 81 (C, γ) pairs.
- The selected pair was $C = 0.01$, $\gamma = 0.001$, the *smoothest* prespecified setting.
- An extended-grid audit continued toward smaller C and inverse bandwidth γ : neither is precisely identified.
- Large γ values fit the training data perfectly but perform near chance on validation data.

The selected RBF boundary is therefore close to linear.



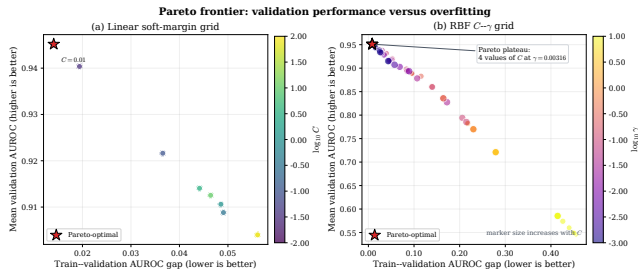
Validation AUROC (left) vs train-validation gap (right).
The star marks the selected smooth model; the top rows overfit.

More flexible RBF settings overfit

Each candidate is plotted by its train-validation gap and validation AUROC. Preferred models are near the **top-left**.

- **Linear** candidates remain close to the top-left.
- **RBF** candidates develop a large train-validation gap as γ increases.
- Four RBF settings share the Pareto frontier at $\gamma = 0.00316$; their differences are numerically negligible.

Key point: The inner-grid frontier is a tuning diagnostic; the untouched outer folds decide whether the kernel helps.



Stars mark Pareto-optimal grid settings. The additional RBF settings widen the overfitting range without improving the final nested comparison.

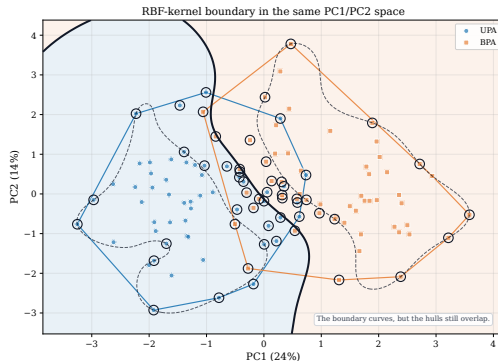
What the RBF boundary looks like

This is the same 2-D projection, now showing the **RBF-kernel** decision surface.

- The boundary is **curved**, not straight.
- It bends around some observations, but the two hulls still overlap.
- A few borderline observations change classification, but the classes remain mixed.

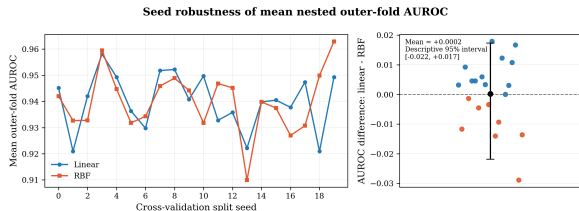
Illustrative 2-D fit (moderate γ); the tuned model chose a much smoother γ .

Key point: A nonlinear boundary changes some local classifications but does not remove the underlying class overlap.



Compared with the linear boundary, the shape changes but the overlap remains.

Does the curved boundary actually help?



We repeated nested cross-validation with 20 different fold assignments and compared mean outer-fold AUROC:

- Mean AUROC: linear 0.9401, RBF 0.9399
- Difference = +0.0002
- Descriptive 95% interval $[-0.022, +0.017]$
- Linear matched or beat RBF in 12 of 20 runs

The same data are reused: this assesses split stability, not independent significance.

Key point: Across repeated splits, the RBF model provides no consistent improvement over the linear model.

The linear soft-margin model is sufficient here

Results

- The hard-margin problem is **infeasible**.
- Linear soft-margin: **20-seed mean AUROC 0.94**.
- RBF kernel: 20-seed mean AUROC 0.94.
- L2-logistic baseline: 20-seed mean AUROC 0.94.

The RBF kernel does not improve mean nested outer-fold performance.

With similar discrimination, the linear model is easier to tune and inspect.

Why it matters

- Hard-margin infeasibility reflects overlap in the feature space.
- Better training fit alone is not a reason to choose a more complex model.
- Because the data are synthetic, the result demonstrates the method rather than clinical performance.

Thank you

Questions?

The numerical analysis is reproducible from the code and synthetic dataset in the project repository.